

Vaishnavi Thirumala

9652335082 | vaishnavithirumala06@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Analytical Artificial Intelligence & Data Science student with experience in applying statistical rigor and explainability to complex machine learning problems. Proficient in designing Bayesian optimization engines, predictive stacking ensembles, and signal-processing pipelines. Skilled at translating high-dimensional data analysis into business-aware, interpretable insights.

EDUCATION

Indian Institute of Information Technology, Sri City
Bachelor of Technology – Artificial Intelligence and Data Science
CGPA: 9.52

Tirupathi, AP
Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: Python, Java, C, SQL
ML/AI: PyTorch, TensorFlow, Scikit-learn, NLP, Reinforcement Learning, Generative AI, XAI (SHAP/LIME)
LLM Frameworks: LangChain, LangGraph, Gemini API, Vector DBs
Data & Tools: Pandas, NumPy, SciPy, MLflow, DVC, Git
Analytics: A/B Testing, Bayesian Inference, Statistical Modeling, Feature Engineering

PROJECTS

ChurnOps: Predictive Modeling Pipeline for Customer Churn

[GitHub](#)

- Engineered a **stacking ensemble** (XGBoost, LightGBM, Random Forest + LR meta-learner via 5-fold CV), improving accuracy from **81.2%** → **90.4%** and ROC-AUC from 0.84 → **0.91** on the IBM Telco dataset (7,043 rows).
- Crafted domain features including *LTV Estimate*, *Bundle Value Density*, and *AvgMonthlySpend*; applied **SHAP** for global and local feature attribution to deliver explainable predictions to business stakeholders.
- Designed a **decision threshold optimizer** evaluating 100 probability intervals against LTV (\$500) and campaign cost (\$50), maximizing net profit to **~\$40,800 per 1,000 customers (560%+ ROI)**.
- Tech:** *XGBoost, LightGBM, Scikit-learn, Optuna, SHAP, Pandera, DVC, MLflow, FastAPI, Evidently AI*

High-Fidelity EEG Compression & Transfer Learning

[GitHub](#)

- Enhanced AVDCT-Net (variational DCT architecture) achieving improved compression ratio (**CR: 8.45**) and reconstruction quality (**QS: 0.80** vs. 0.46 baseline), verified against the BCI Competition II dataset, BCI Competition III dataset, Bonn Dataset.
- Engineered cross-subject **transfer learning** pipeline to mitigate EEG non-stationarity and inter-subject variability, significantly improving generalization across subjects.
- Developed **DSP preprocessing** pipeline using Butterworth band-pass filtering (0.5–30 Hz) and channel-wise normalization for 64-channel EEG signals to reduce noise and extract clean spectral features.
- Tech:** *PyTorch, VAE, DSP (SciPy), Butterworth Filters, BCI Competition III Dataset, NumPy, Matplotlib*

Fake News Classification using NLP

[GitHub](#)

- Engineered a robust text preprocessing pipeline using **NLTK** for lemmatization, stop-word removal, and RegEx cleaning on the WELFake dataset (72k+ articles).
- Implemented **TF-IDF Vectorization** with uni-gram to tri-gram extraction; optimized **Logistic Regression** and **Multinomial Naive Bayes** via **GridSearchCV**, achieving a peak **F1-Score of 0.90**.
- Conducted comprehensive evaluation using confusion matrices and precision-recall analysis to validate classifier robustness on imbalanced real-vs-fake distributions.
- Tech:** *Python, Scikit-learn, NLTK, TF-IDF, GridSearchCV, Pandas, Matplotlib, Seaborn*

LEADERSHIP & EXPERIENCE

NLP Domain Lead
Epoch – AI/ML Club of IIIT Sri City
Design Team Core Member
IIIT Sri City

Aug 2025 – April 2026
Sri City, AP
Oct 2024 – Aug 2025
Sri City, AP